Higher National Diploma in Engineering (Civil/Mechanical)

Higher National Diploma in Building Services Engineering

First Year, 2nd Semester Examination-2022

ME 1206/BSE 1205 - Fundamentals of Thermodynamics

answer any five questions. Time : 3 hours

For air, $R = 0.287\ \mathrm{kJ/kg\ K}$, $C_p = 1.005\ \mathrm{kJ/kg\ K}$ or $C_v = 0.718\ \mathrm{kJ/kg\ K}$.

Question 01

i. State the **first law of Thermodynamics**. (05 marks) ii. State the **non-flow energy equation**. (04 marks) iii. A gas with a volume of $0.09\ \mathrm{m}^3$, pressure of $2.75 \times 10^5\ \mathrm{N/m}^2$ and temperature of $185^\circ C$ is compressed at **constant pressure** until its temperature becomes $15^\circ C$. Calculate: (a) Mass of the gas. (04 marks) (b) Heat transferred. (04 marks) (c) Work done. (04 marks) (d) Internal energy change. (04 marks)

Question 02

i. Define following terms related to pure substances: (06 marks) (a) **Dryness fraction**. (b) **Saturation temperature**. (c) **Superheated vapor**. ii. Verify an equation to calculate the **specific volume of wet vapor** by obtaining a relationship with the dryness fraction. (04 marks) iii. Steam enters a Turbine at a pressure $15\ \mathrm{bar\ absolute}$ and $250^\circ C$. It is exhausted at $1.5\ \mathrm{bar}$ having a specific volume of $0.8113\ \mathrm{m}^3/\mathrm{kg}$. If the mass flowrate of steam is $10\ \mathrm{kg/s}$. Find: (a) Dryness fraction of steam at leaving. (15 marks total for a, b, c) (b) Drop in enthalpy. (c) Change in internal energy.

Question 03

i. Obtain equations for the following thermodynamic systems applying the **steady flow energy equation**. Write all the assumptions you make. (10 marks) (a) Work done on a **Compressor**. (b) Heat supply to a **Boiler**. ii. In a gas turbine unit, the gases flow through the turbine at a rate of $15\ \mathrm{kg/s}$ and the power developed by the turbine is $12000\ \mathrm{kW}$. The enthalpies of gases at the inlet and outlet are $1260\ \mathrm{kJ/kg}$ and $400\ \mathrm{kJ/kg}$ respectively. The velocity of gases at the inlet and outlet are $50\ \mathrm{m/s}$ and $110\ \mathrm{m/s}$ respectively. Calculate: (15 marks) (a) Rate at which **heat is rejected to the turbine**. (b) **Area of the inlet pipe** given that the specific volume of the gases at the inlet is $0.45\ \mathrm{m^3/kg}$.

Question 04

i. Briefly describe the operation of following: (09 marks) (a) **Heat engine**. (b) **Heat pump**. (c) **Refrigerator**. ii. Explain in brief what is a **reversible thermodynamic cycle**. (04 marks) iii. An engine works between the temperature limits of $2000\ \mathrm{K}$ and $400\ \mathrm{K}$. (12 marks) (a) What can be the **maximum thermal efficiency** of the engine? (b) Find the **maximum work produced** by the engine if it extracts $1050\ \mathrm{kJ}$ of heat energy from the heat source. (c) If a heat pump works between the same temperature limits, what is the **Coefficient of Performance** of it?

Question 05

i. Name **three modes of heat transfer** giving two applications for each. (06 marks) ii. State the **Fourier's Law of heat transfer**. (04 marks) iii. A fire occurs inside a room and the inner surface of the plane brick wall is at $85^\circ C$ and the outer surface is at $40^\circ C$. Calculate the **rate of heat transfer** if surface area of the wall is $45\ \mathrm{m^2}$, which is $150\ \mathrm{mm}$ thick. The thermal conductivity of the brick is $0.51\ \mathrm{W/m^\circ C}$. (06 marks) iv. State the **Stefan-Boltzmann Law of Thermal Radiation**. (04 marks) v. If the surface temperature of a metal spherical ball is $150^\circ C$ and the radius is $0.1\ \mathrm{m}$, what is the **power emitted by ball** if it is treated as a **black body**? Stefan-Boltzmann constant is $5.67 \times 10^{-8}\ \mathrm{W/m^2K^4}$. (05 marks)

Saturated Water and Steam

p [bar]	T_s [°C]	v_g [m³/kg]	u_f [kJ/kg]	u_g [kJ/kg]	h_f [kJ/kg]	h_{fg} [kJ/kg]	h_g [kJ/kg]	s_f [kJ/kg K]	s_{fg} [kJ/kg K]	s_g [kJ/kg K]
1.0	99.6	1.694	417	2506	417	2258	2675	1.303	6.056	7.359
1.1	102.3	1.549	429	2510	429	2251	2680	1.333	5.994	7.327
1.2	104.8	1.428	439	2512	439	2244	2683	1.361	5.937	7.298
1.3	107.1	1.325	449	2515	449	2238	2687	1.387	5.884	7.271
1.4	109.3	1.236	458	2517	458	2232	2690	1.411	5.835	7.246
1.5	111.4	1.159	467	2519	467	2226	2693	1.434	5.789	7.223
1.6	113.3	1.091	475	2521	475	2221	2696	1.455	5.747	7.202
1.7	115.2	1.031	483	2524	483	2216	2699	1.475	5.707	7.182
1.8	116.9	0.9774	491	2526	491	2211	2702	1.494	5.669	7.163
1.9	118.6	0.9292	498	2528	498	2206	2704	1.513	5.632	7.145
2.0	120.2	0.8856	505	2530	505	2202	2707	1.530	5.597	7.127
2.1	121.8	0.8461	511	2531	511	2198	2709	1.547	5.564	7.111
2.2	123.3	0.8100	518	2533	518	2193	2711	1.563	5.533	7.096
2.3	124.7	0.7770	524	2534	524	2189	2713	1.578	5.503	7.081
2.4	126.1	0.7466	530	2536	530	2185	2715	1.593	5.474	7.067
2.5	127.4	0.7186	535	2537	535	2182	2717	1.607	5.446	7.053

Superheated Steam*

p/[bar] (T_s/[°C])			T [°C]	200	250	300	350	400	450	500	600
5 (151.8)	v_g	0.3748	v	0.4252	0.4745	0.5226	0.5701	0.6172	0.6641	0.7108	0.8040
	u_g	2562	u	2644	2725	2804	2883	2963	3045	3129	3300
	h_g	2749	h	2857	2962	3065	3168	3272	3377	3484	3702
	s_g	6.822	s	7.060	7.271	7.460	7.633	7.793	7.944	8.087	8.351
6 (158.8)	v_g	0.3156	v	0.3522	0.3940	0.4344	0.4743	0.5136	0.5528	0.5919	0.6697
	u_g	2568	u	2640	2722	2801	2881	2962	3044	3128	3299
	h_g	2757	h	2851	2958	3062	3166	3270	3376	3483	3701
	s_g	6.761	s	6.968	7.182	7.373	7.546	7.707	7.858	8.001	8.267
7 (165.0)	v_g	0.2728	v	0.3001	0.3364	0.3714	0.4058	0.4397	0.4734	0.5069	0.5737
	u_g	2573	u	2636	2720	2800	2880	2961	3043	3127	3298
	h_g	2764	h	2846	2955	3060	3164	3269	3374	3482	3700
	s_g	6.709	s	6.888	7.106	7.298	7.473	7.634	7.786	7.929	8.195
8 (170.4)	v_g	0.2403	v	0.2610	0.2933	0.3242	0.3544	0.3842	0.4138	0.4432	0.5018
	u_g	2577	u	2631	2716	2798	2878	2960	3042	3126	3298
	h_g	2769	h	2840	2951	3057	3162	3267	3373	3481	3699
	s_g	6.663	s	6.817	7.040	7.233	7.409	7.571	7.723	7.866	8.132
9 (175.4)	v_g	0.2149	v	0.2305	0.2597	0.2874	0.3144	0.3410	0.3674	0.3937	0.4458
	u_g	2581	u	2628	2714	2796	2877	2959	3041	3126	3298
	h_g	2774	h	2835	2948	3055	3160	3266	3372	3480	3699
	s_g	6.623	s	6.753	6.980	7.176	7.352	7.515	7.667	7.811	8.077
10 (179.9)	v_g	0.1944	v	0.2061	0.2328	0.2580	0.2825	0.3065	0.3303	0.3540	0.4010
	u_g	2584	u	2623	2711	2794	2875	2957	3040	3124	3297
	h_g	2778	h	2829	2944	3052	3158	3264	3370	3478	3698
	s_g	6.586	s	6.695	6.926	7.124	7.301	7.464	7.617	7.761	8.028
15 (198.3)	v_g	0.1317	v	0.1324	0.1520	0.1697	0.1865	0.2029	0.2191	0.2351	0.2667
	u_g	2595	u	2597	2697	2784	2868	2952	3035	3120	3294
	h_g	2792	h	2796	2925	3039	3148	3256	3364	3473	3694
	s_g	6.445	s	6.452	6.711	6.919	7.102	7.268	7.423	7.569	7.838